Prenatal Alcohol and Fetal Alcohol Spectrum Disorders

Prenatal alcohol is causally implicated in FASD spanning growth, facial, and neurobehavioral outcomes; detected exposure is not equivalent to FASD, and not all neurobehavioral problems are alcohol-caused.

GRADE: HIGH-FASD; LOW-ANY-SIP | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Prenatal alcohol exposure is causally implicated in a clinical continuum (FASD) of growth restriction, characteristic facial features, and neurobehavioral impairment. Severity is dose- and timing-dependent. A detected exposure is not a diagnosis; a neurobehavioral problem is not proof of alcohol causation.

EVIDENCE AND CAUSAL ASSESSMENT

Hoyme et al. (2016) update diagnostic criteria from evaluation of more than 10,000 children and NIAAA-linked epidemiology (CIFASD, CoFASP): documented prenatal alcohol exposure, facial features, growth, CNS, and neurobehavior. This is clinical teratology and consensus diagnosis, not a randomized exposure trial. Human causal inference rests on dose, pattern, dysmorphology, and exclusion of competing causes.

EFFECT SIZE AND PERSISTENCE

FASD persistence is often substantial. Unmeasured low-level exposure does not inherit that effect size.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (teratological toxicology, not an observational population phenotype association).

Exposure under-report; polysubstance use; SES and nutrition; stigma; diagnostic substitution.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean every drink produces FASD, or that every developmental difficulty is alcohol-caused, or that affected children have a fixed fate.

WHAT ACTION IS REASONABLY JUSTIFIED

Avoiding alcohol in pregnancy and supporting affected children are justified. Shame as a developmental theory is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Hoyme et al. (2016), Pediatrics